

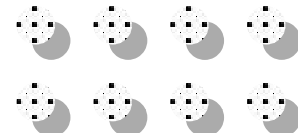
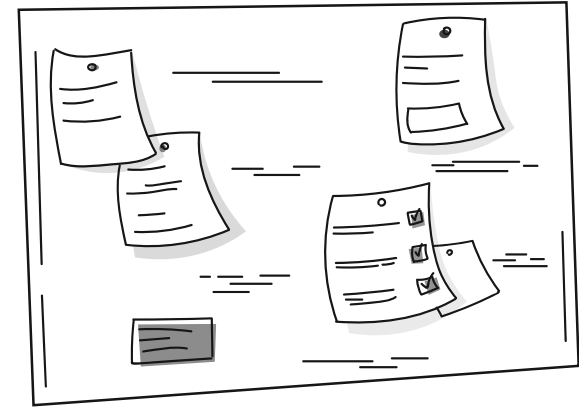
Hands-On Session 1 – 2 - 3

Machine & Deep Learning Applications in Aquatic Organisms' Multi-Omic Data

25-10-21

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 - Deep learning shortcomings in microbiome
- Session 3 – Pipelines, Data2Interpretation
 - 16S Targeted Amplicon Sequencing – Pipelines
 - 16S Targeted Amplicon Sequencing – An outline
 - Whole Metagenome Shotgun Sequencing – Pipelines
 - Whole Metagenome Shotgun Sequencing – An outline
- Take home...

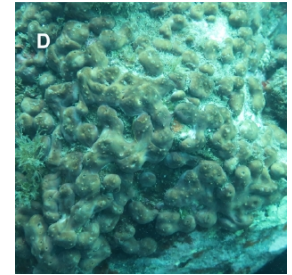
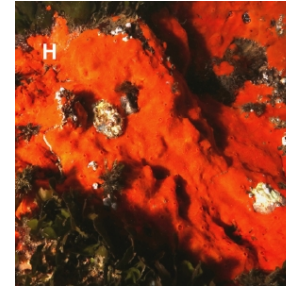
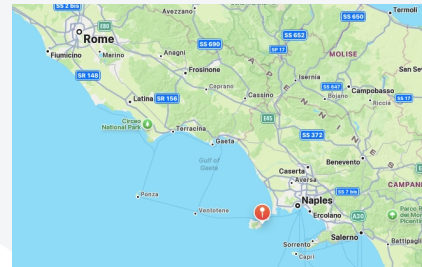
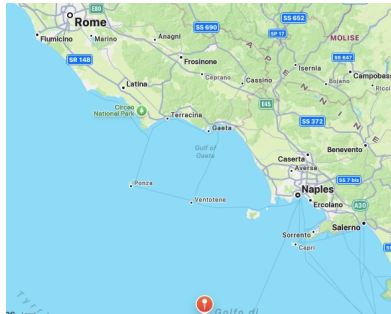


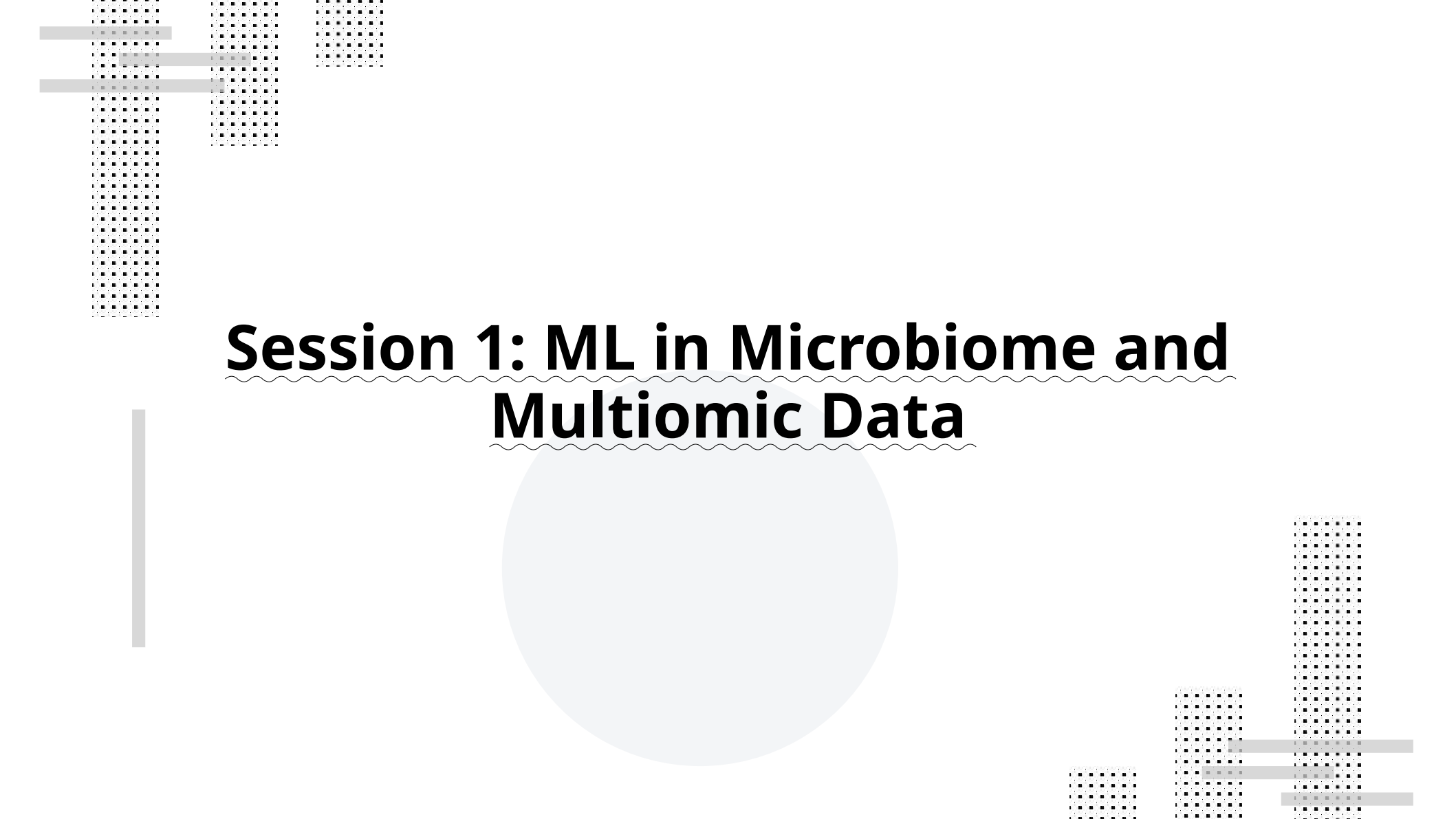
Chosen Data

<u>Species</u>	<u>Number of Samples</u>	<u>Site</u>
C. crambe	5-2	GF-SA
C. nucula	5-5	GF-SA
C. reniformis	5-5	GF-SA
P. ficiformis	3-5	GF-SA

GF: Grotta Mago

SA: Sant'Anna



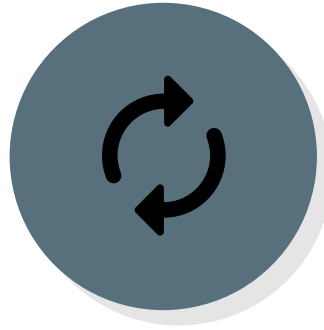


Session 1: ML in Microbiome and Multiomic Data

General Look at AI in Microbiome



- ML-DL workflow:
 - Biological question
 - Setting – classification, regression, clustering...
 - Integration level



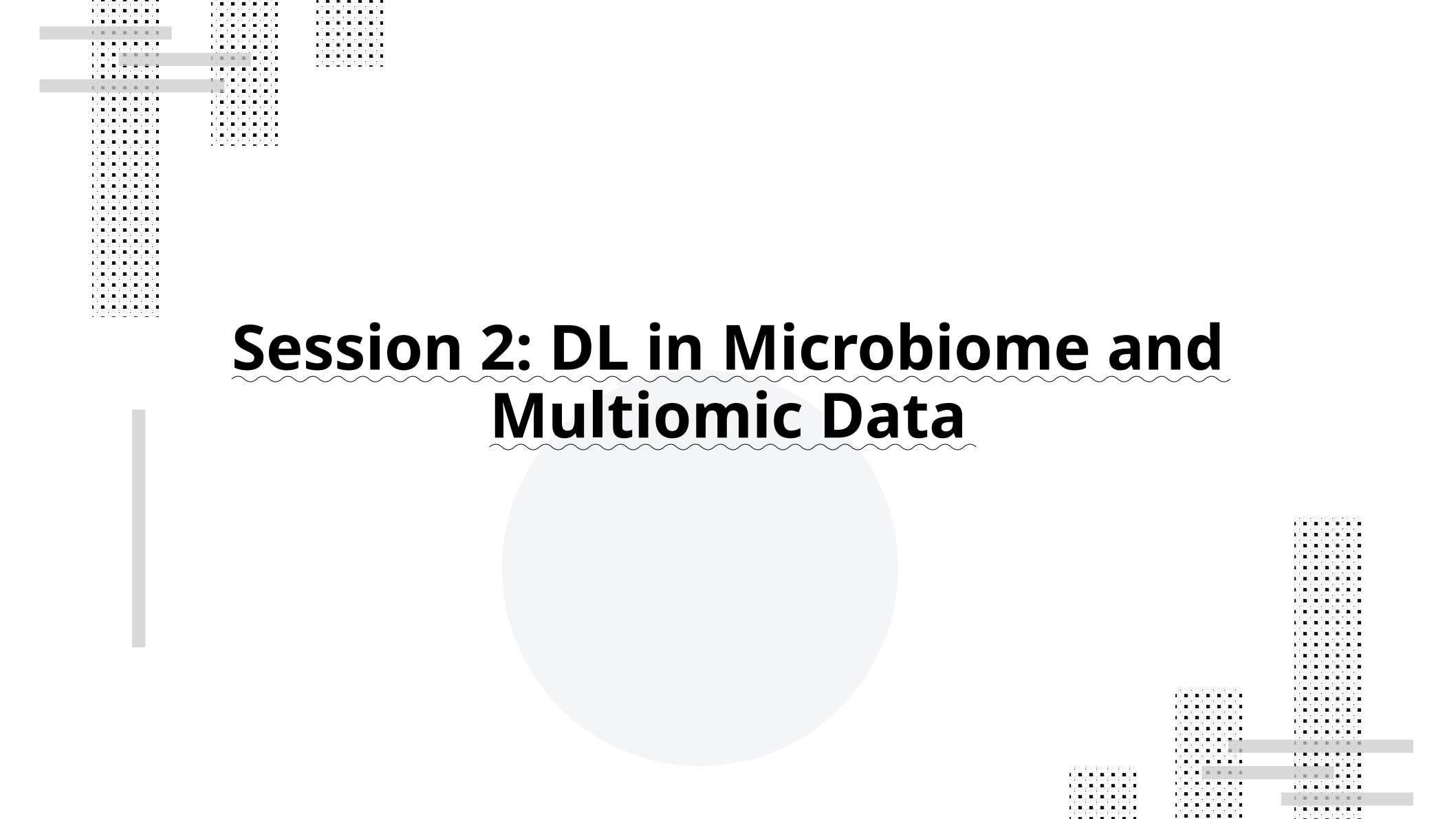
- QC:
 - ROC-AUC – test & training sets
 - PR-AUC
 - Accuracy – Error rate



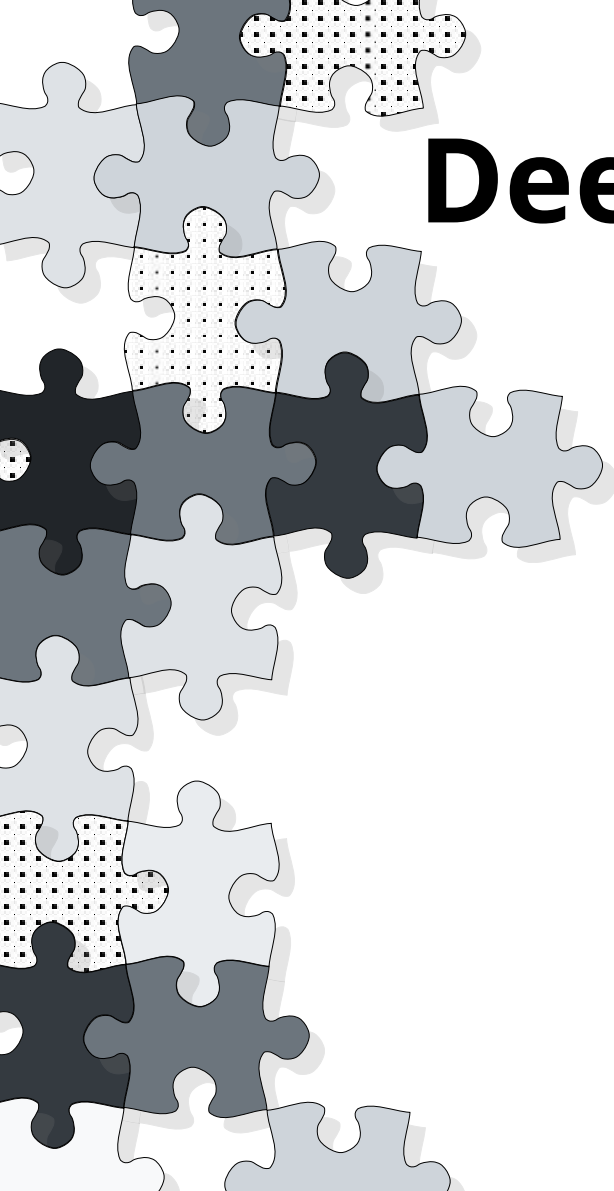
- Obtain:
 - Filter down possibilities
 - Potential biomarkers
 - Effect sizes

AI in Microbiome Studies

Disease detection	CNN/autoencoder	Oh, M., & Zhang, L. (2020)
Large scale annotation	large-sized omic integration with ML annotation	Carradec, Q. et al (2018)
Microbial community and temporal dynamics prediction	wastewater treatment plants	Andersen, K. S. <i>et al</i> (2025)
AI based feeding technique - Shrimp biomass detection	Image-based deep learning analysis	Darapaneni, N. <i>et al</i> (2022)
Image-based disease detection	SVM (Support Vector Machines)	Chen F. <i>et al</i> (2022)
Water quality prediction	artificial neural networks	Chen, Y. <i>et al</i> (2020)
Fish species classification	Multi-class SVM	Hu et al (2012)



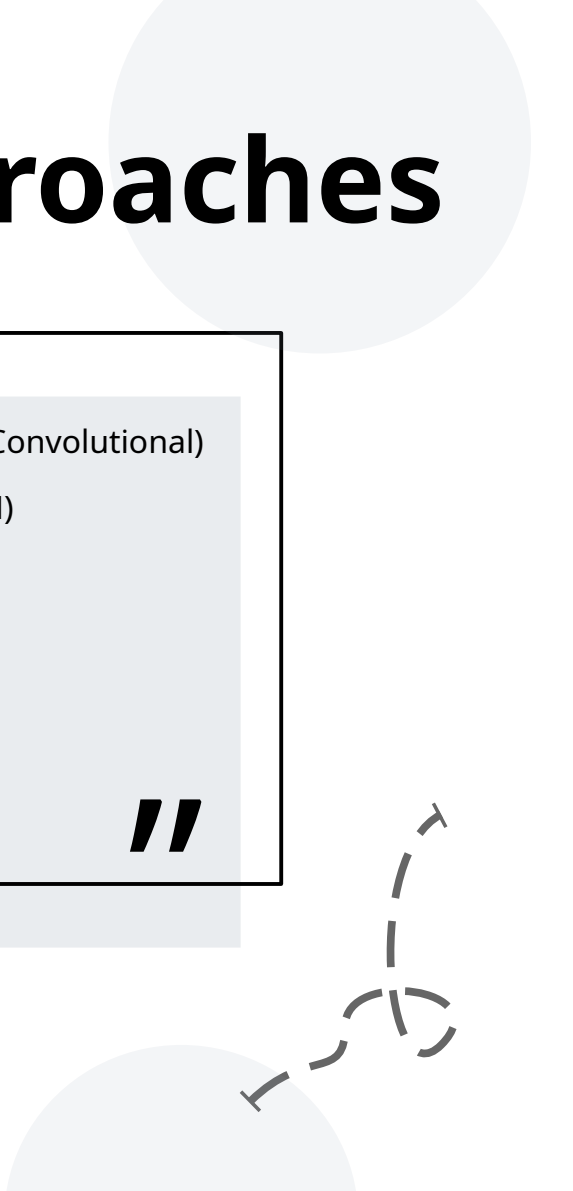
Session 2: DL in Microbiome and Multiomic Data



Deep Learning Approaches

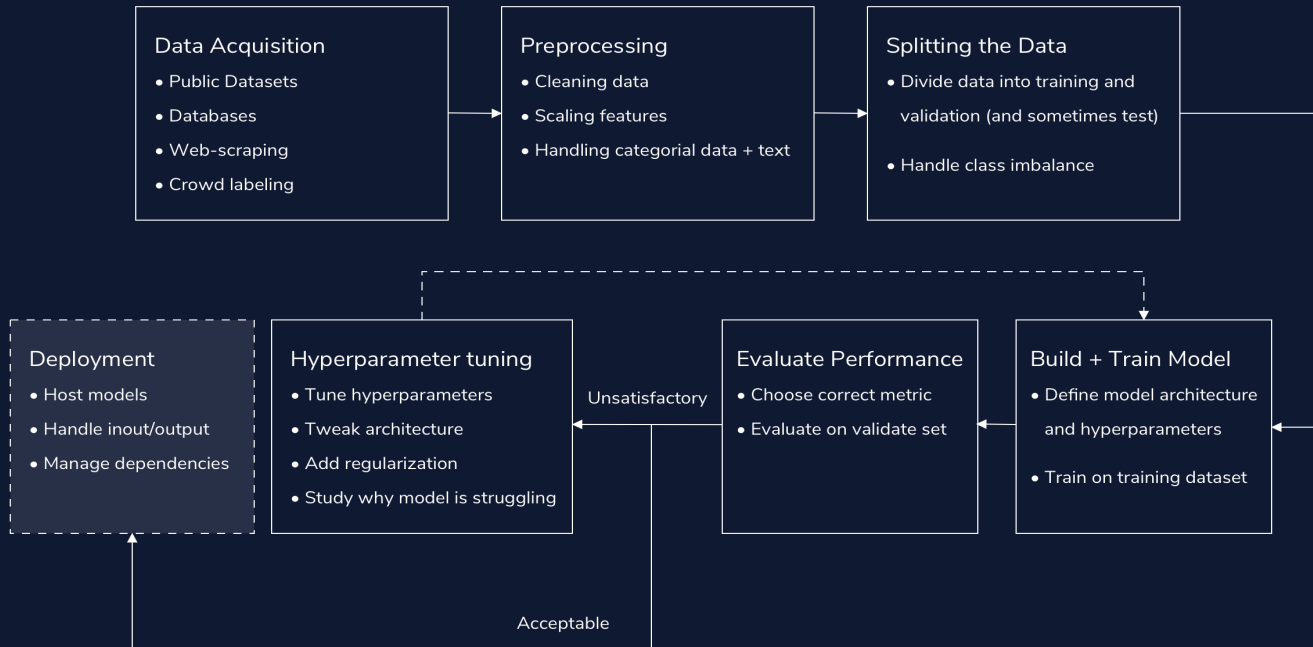
// Neural Networks (Recurrent, Graph and Convolutional)
Long Short-Term Memory Network (LSTM)
Variational Auto Encoders
Neural Matrix Factorization
SHAP/Integrated Gradients*
DIABLO (MixOmics)**
Similarity Network Fusion Method**

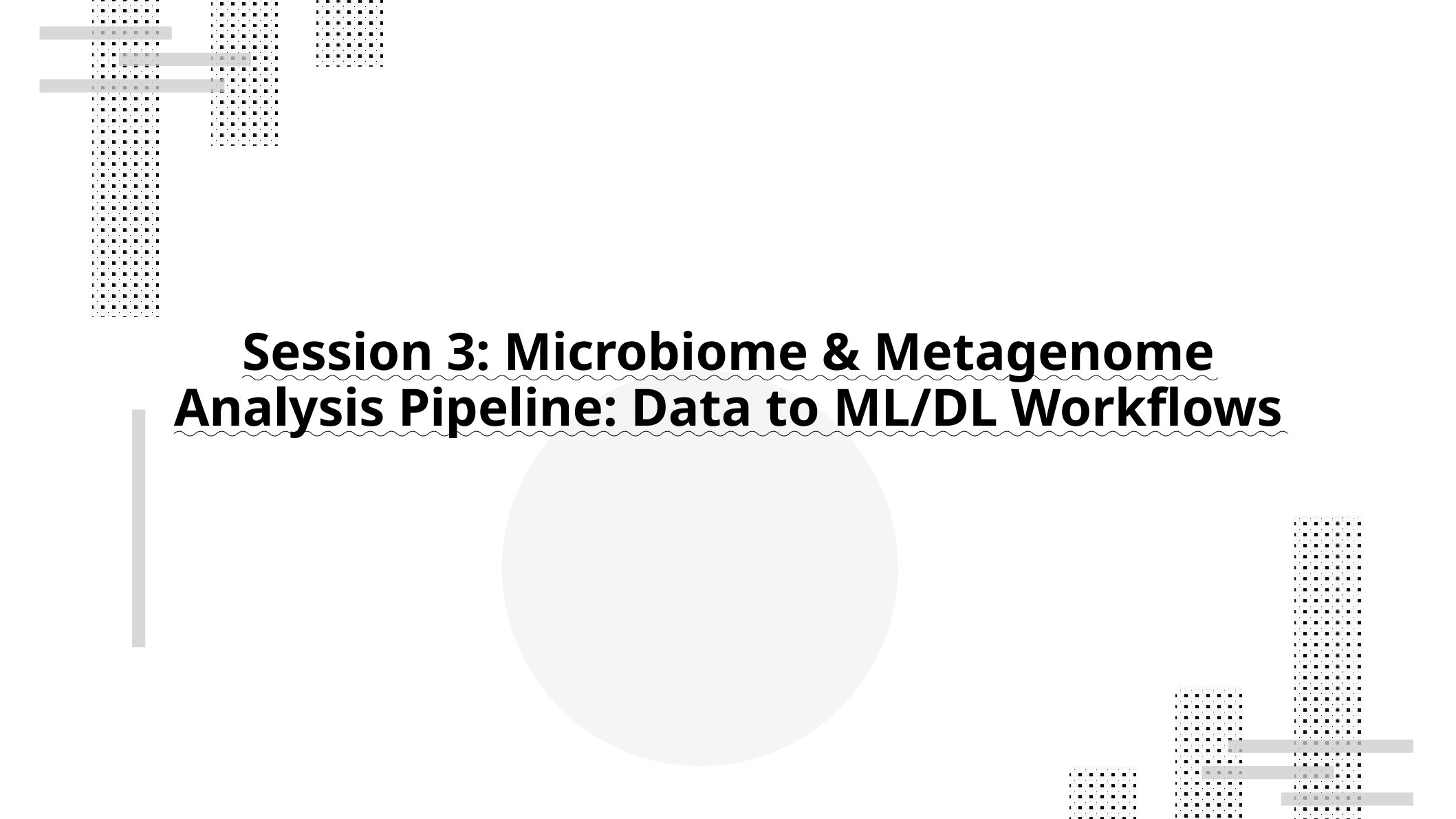
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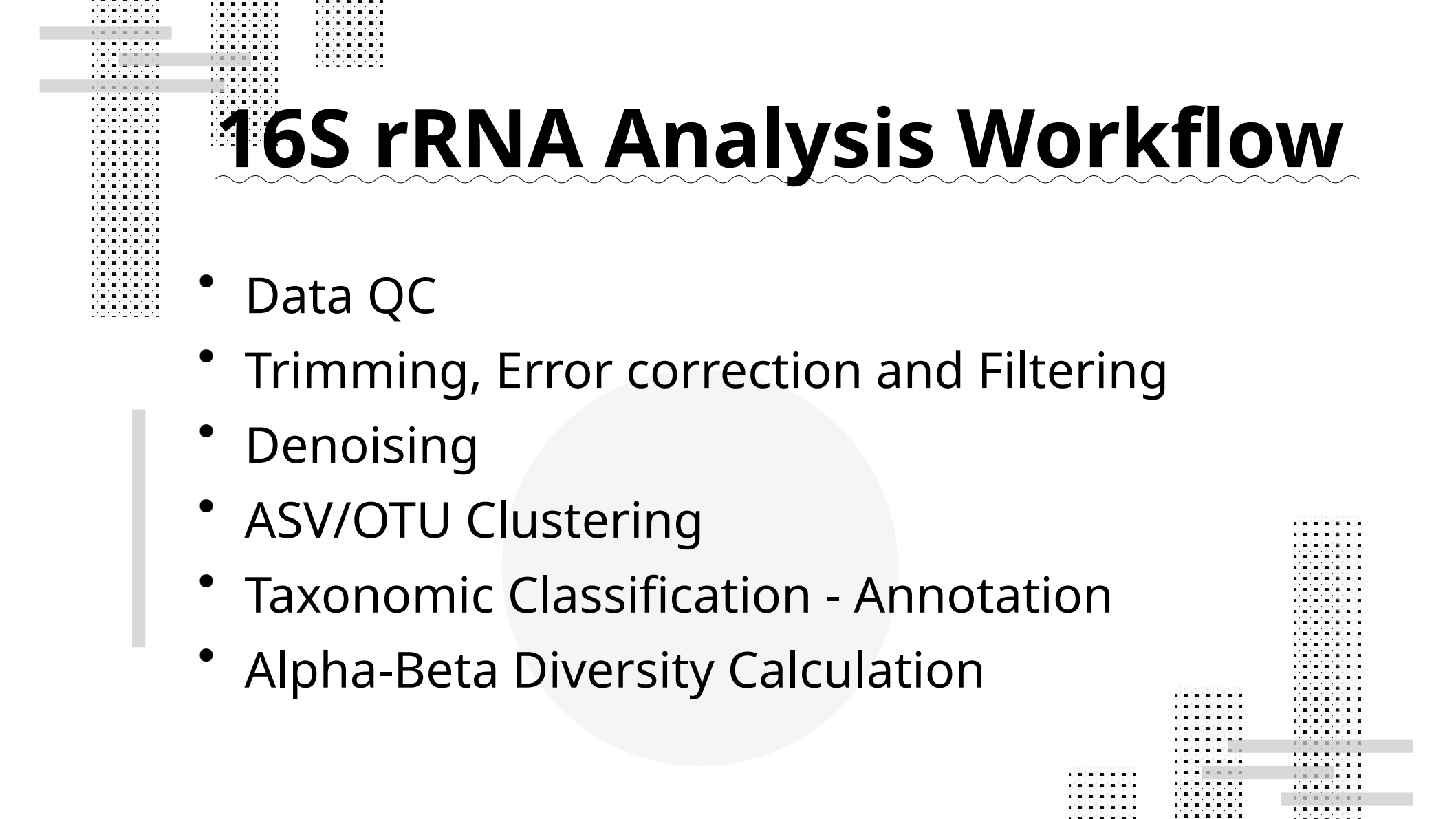
Session 2: Deep Learning Workflows, Microbiome & Multiomics

- Sparse data
- Few samples per study
- Biological noise
- High dimensionality
- Different scales noise, and feature spaces
- Lack of standardization
- Biological bias and knowledge gaps
- Mislabels
- Data leakage





Session 3: Microbiome & Metagenome Analysis Pipeline: Data to ML/DL Workflows

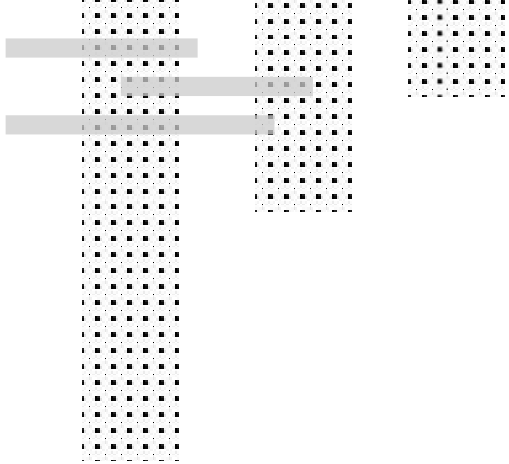


16S rRNA Analysis Workflow

- Data QC
- Trimming, Error correction and Filtering
- Denoising
- ASV/OTU Clustering
- Taxonomic Classification - Annotation
- Alpha-Beta Diversity Calculation

WMS Analysis Workflow

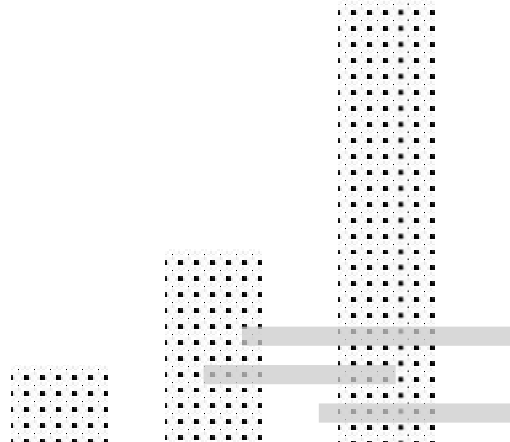
- Data QC
- Trimming, Error Correction
- Host filtering
- MAG Creation
- MAG QC
- MAG Coverage – Abundance Calculation
- MAG Binning
- MAG Bin QC
- Gene Annotation
- Short read Taxonomic Classification
- Functional Annotation
- PCA-PcoA-ML-DL Workflows



Thank you for listening...



Any
questions?



References

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